

The Self: 0/0 of Identity

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The Removable Singularity Framework

Abstract

The Ship of Theseus (Plutarch): if every part is replaced, is it the same ship? We show identity IS a 0/0. ~98% of the body's atoms are replaced each year; after ~10 years essentially zero original matter remains, yet "you" persist. In Game of Life the glider (Ch.54) is the perfect metaphor: the pattern persists while the cells are replaced (overlap reaches 0/5 yet the glider continues). The no-cloning theorem (Wootters-Zurek 1982) shows the self cannot be duplicated, only continued (max gap 0.5858). Identity is information flow, not substance; the self is a removable singularity of the body.

1. The Atom Turnover

$$\text{original_after_years} = (0.02)^{\text{years}}$$

About 98% of the body's atoms are replaced each year. Computed: after 1 year $2.0e-02$ remain; after 10 years $1.0e-17$ - ZERO original matter. Yet you continue: the PATTERN persists, not the parts. (Cells turn over; ~98% noted in broad cell-biology reviews.)

2. The Life Metaphor: A Glider Is a Self

A glider in Game of Life (Ch.54) is an information pattern moving through a grid. Computed: its 5 identity cells are replaced rapidly - original live sites drop to 0/5 within ~8 generations - yet the glider (the pattern) persists all 16 generations. THE GLIDER IS THE SHIP OF THESEUS: the self is the moving pattern, not the traced cells.

3. The Ship of Theseus

Computed: 1000 planks, 4 replaced per day. Material overlap (original planks): day 0 = 1.000, day 125 = 0.500, day 200 = 0.200, day 250 = 0.000. Yet the ship sailed every day - the pattern and the name continued. The identity function is a 0/0: it never empties at the cutoff, because it lives in the pattern.

4. The No-Cloning Theorem (Wootters-Zurek 1982)

An unknown quantum state cannot be duplicated. A universal cloner $|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle$ violates linearity: linear evolution forces $\alpha|00\rangle + \beta|11\rangle$ while a perfect copy requires $\alpha^2|00\rangle + \alpha\beta(|01\rangle + |10\rangle) + \beta^2|11\rangle$. Computed gap reaches 0.5858 at $\alpha=1/\sqrt{2}$. So the self cannot be copy-pasted - only CONTINUED or destroyed. Immortality is pattern continuation, not duplication.

5. Immortality in the Simulation

If consciousness = integrated information (Ch.53) and the universe = information (Ch.52), then the self = a persistent information pattern. An uploaded mind is a pattern that persists as long as the simulation runs (Ch.54; Chalmers 2010). Death and life meet at the boundary of the pattern - the same event seen from inside or outside. The 0/0 of birth and death.

6. Connections

Concept	Prior Chapter	Connection
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Identity	Ch.54 Simulation	<i>Puno (2026) - The Self: 0/0 of Identity</i> We are gliders on the grid
Mind	Ch.53 Hard Problem	Self as integrated Phi
Agency	Ch.55 Free Will	The self that chooses
Information	Ch.52 Info paradox	Identity is information
Quantum	Ch.31 Quantum	No-cloning: one self only
Time	Ch.48 Arrow of time	The pattern flows forward

7. Conclusion

The self IS a 0/0: the boundary where molecules become "I". No enduring particle exists; identity is an information flow through matter. The Ship of Theseus is resolved: the ship is the pattern. The glider persists while its cells turn over; the body renews while the self continues; no-cloning means we are continued, never copied. Identity is pattern over substance - the removable singularity of the body.

References

- [1] Plutarch, Life of Theseus: the Ship of Theseus
- [2] Wootters & Zurek (1982), no-cloning theorem
- [3] Chalmers (2010), The Singularity: upload and the mind
- [4] Puno, "The Removable Singularity" (2026)